Analysis based on the Gasparrini article.

Case studies Barranquilla, Cartagena and Santa Marta

—

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Background

Hablar de la tesis de Diego y los datos satelitales

Preparation

The following packages are loaded in the session, and need to be installed to run the R code:

```
library(readr); library(dlnm) ; library(gnm) ; library(data.table) ; library(splines);
library(MASS);  library(ggcorrplot); library(dplyr); library(car);library(tidyr);
library(lme4);  library(kableExtra)
```

```
Datosdengueclima <- read_csv("data/caribe_data/Datosdengueclimav.csv")
```

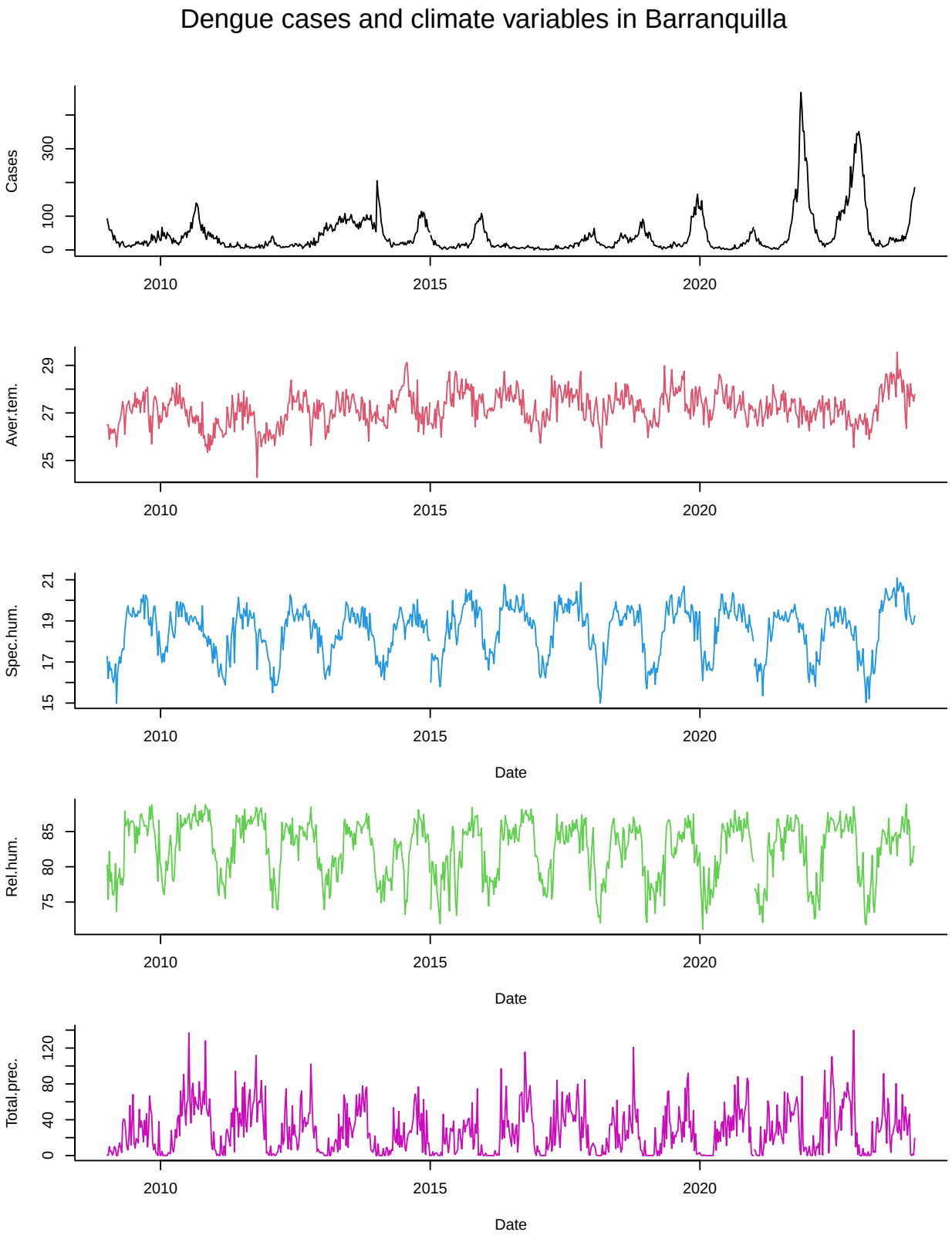
```
## New names:
## Rows: 807 Columns: 33
## -- Column specification
## -------------------------------------------------------- Delimiter: "," dbl
## (33): ...1, Año, Semana_Epidemiologica, Casos_Barranquilla, Casos_Cartag...
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

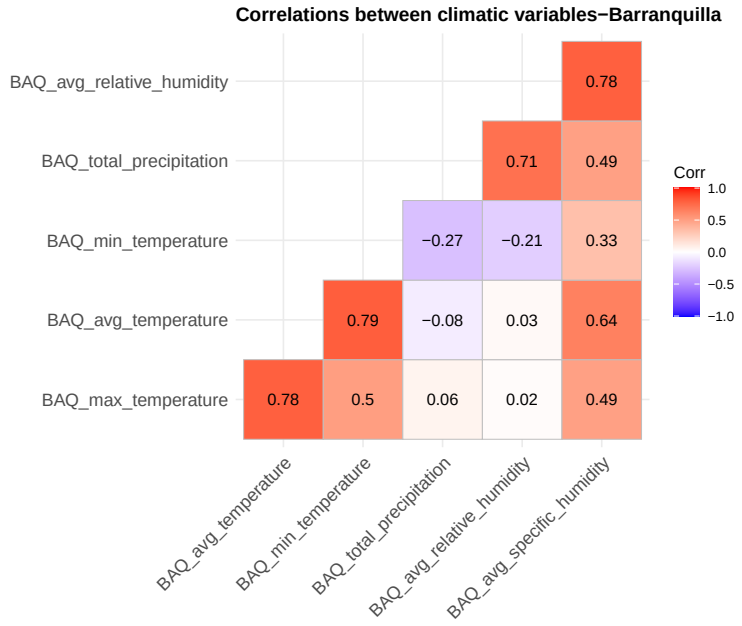
```
datos_dlnm=Datosdengueclima
datos_dlnm <- datos_dlnm[datos_dlnm$Año >= 2009 & datos_dlnm$Año <= 2023, ]
datos_dlnm$Fecha <- as.Date(paste(datos_dlnm$Año, datos_dlnm$Semana_Epidemiologica, 1,
                                  sep = "-"),
                            format = "%Y-%U-%u")
```

Analysis

Barranquila

The archive variables for each city during the study period are illustrated below:


Dengue cases and climate variables in Barranquilla
Cases
0
100
300
2010
2015
2020
Aver.tem.
25
27
29
2010
2015
2020
Spec.hum.
15
17
19
21
2010
2015
2020
Date
Rel.hum.
75
80
85
2010
2015
2020
Date
Total.prec.
0
40
80
120
2010
2015
2020
Date



Now, with the function `crossbasis()` a time-distributed transformation is parameterised to study the relationship between environmental exposures (temperature, precipitation and humidity) and dengue cases.

```
cb_temp_ave_bql<- crossbasis(
                  datos_dlnm$BAQ_avg_temperature,
                  lag = 24, # Up to 24 weeks lags
                  argvar = list(fun = "ns", df = 3), # Non-linear relationship with temperature
                  arglag = list(fun = "poly", degree = 2) # Non-linear relationship with lags
                 )

cb_hum_rel_bql<- crossbasis(
                  datos_dlnm$BAQ_avg_relative_humidity,
                  lag = 24,
                  argvar = list(fun = "ns", df = 3),
                  arglag = list(fun = "poly", degree = 2)
                 )

cb_prec_tot_bql<- crossbasis(
                  datos_dlnm$BAQ_total_precipitation,
                  lag = 24,
                  argvar = list(fun = "ns", df = 3),
                  arglag = list(fun = "poly", degree = 2)
                 )
```

Now, with all terms parameterised, a negative binomial regression is fitted excluding variables that were highly correlated, as follows:

```
mod_bql <- glm.nb(Casos_Barranquilla ~ cb_temp_ave_bql +
                                       cb_hum_rel_bql  +
                                       cb_prec_tot_bql+
                                       ns(Semana_Epidemiologica, df = 7) + # Seasonal adjustment
                                       factor(Año), # Trend by year
                                      data=datos_dlnm)
```

```r
vif(mod_bql)
```

```
##                                       GVIF Df GVIF^(1/(2*Df))
## cb_temp_ave_bql                    36342.39  9        1.792076
## cb_hum_rel_bql                   4460330.39  9        2.341042
## cb_prec_tot_bql                  2058664.82  9        2.242615
## ns(Semana_Epidemiologica, df = 7)  114092.11  7        2.297378
## factor(Año)                          4311.03 14        1.348362
```

Now, with `crosspred()`, we take the results of the model (the coefficients) and evaluate how these variables are related to dengue risk in different scenarios:

```r
cptmean <- crosspred(
           basis = cb_temp_ave_bql,
           model = mod_bql,
           cen = mean(datos_dlnm$BAQ_avg_temperature, na.rm = TRUE),
           by = 1)

cphrh <- crosspred(
         basis = cb_hum_rel_bql,
         model = mod_bql,
         cen = mean(datos_dlnm$BAQ_avg_relative_humidity, na.rm = TRUE),
         by = 0.5)

cpprecip <- crosspred(
            basis = cb_prec_tot_bql,
            model = mod_bql,
            cen = mean(datos_dlnm$BAQ_total_precipitation, na.rm = TRUE),
            by = 5 )
```

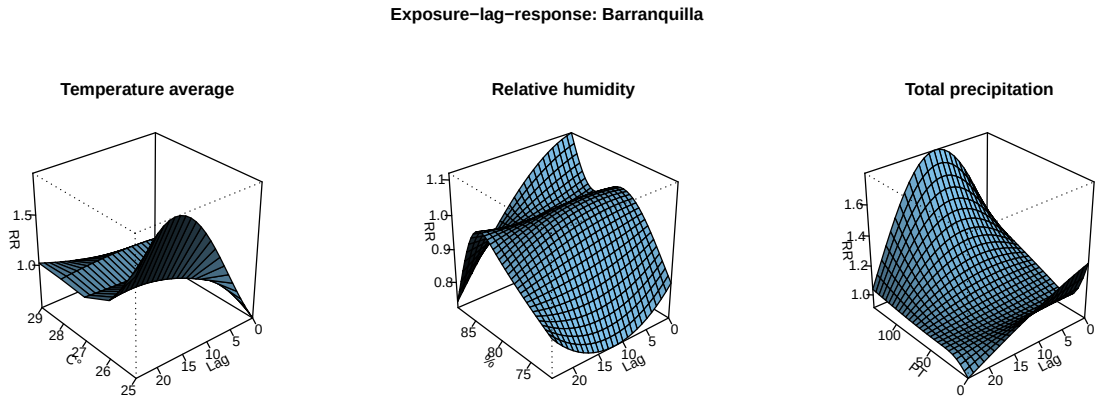
With the above, the association in both dimensions of exposure intensity and delay is plotted. Specifically, the following graphs represent the exposure responses (interpreted as the net associations taking into account the whole lag period), the full two-dimensional exposure-delay responses for non-linear temperature, humidity, precipitation and the lag response.

```r
par(oma = c(0, 0, 5, 0))

plot(cphrh,
     xlab = "%",
     zlab = "RR",
     main = "Relative humidity",
     phi = 30, theta = 230)

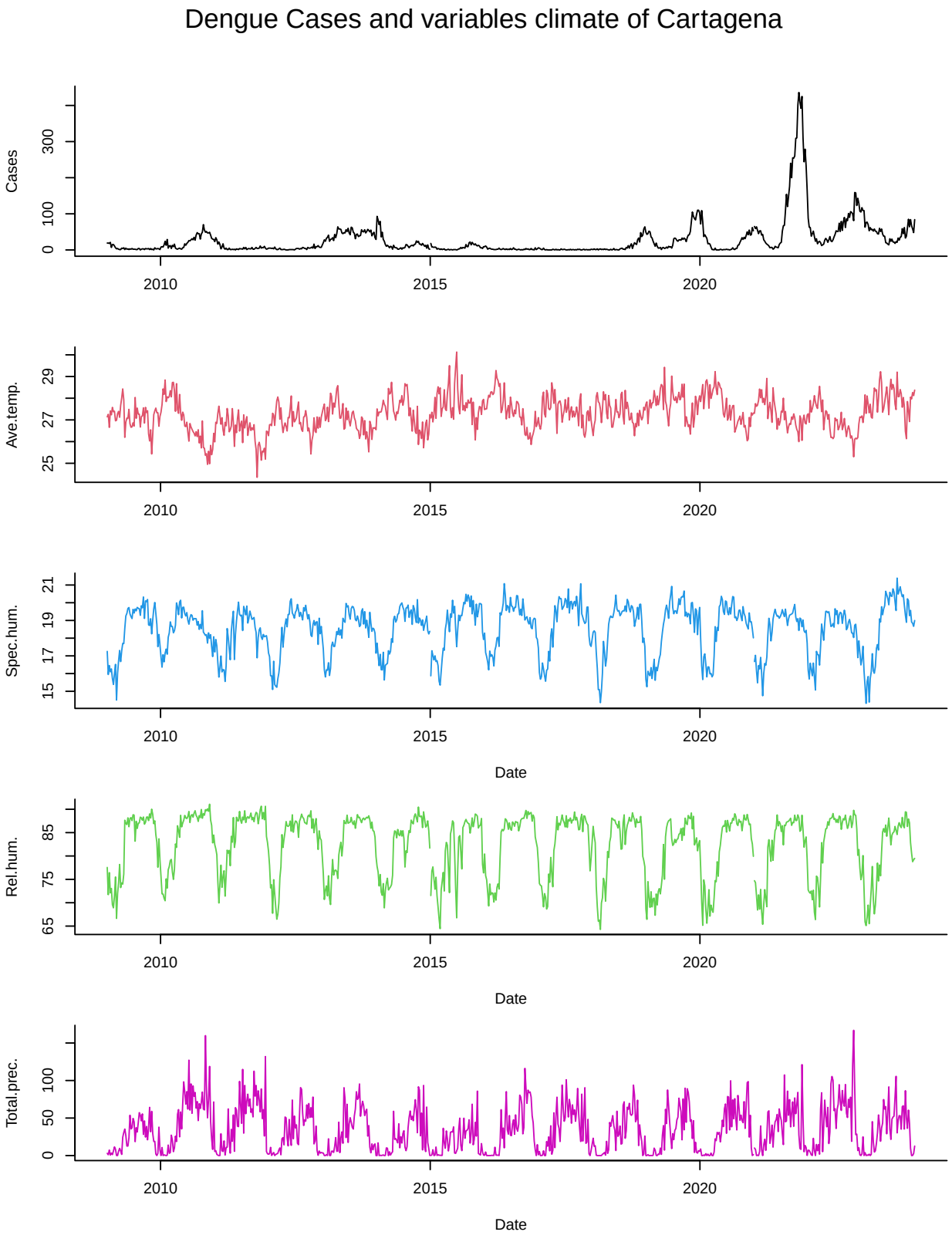
plot(cpprecip,
     xlab = "PT",
     zlab = "RR",
     main = "Total precipitation",
     phi = 30, theta = 230)

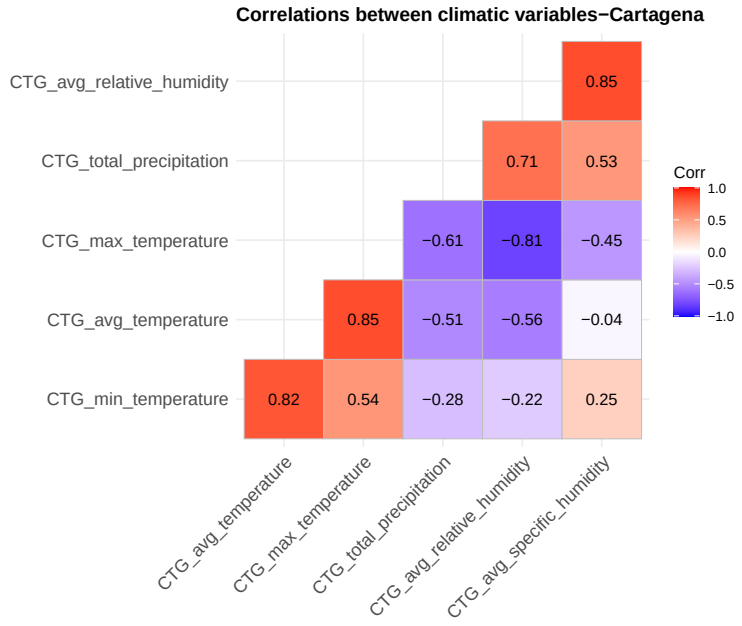
title(main = "Exposure-lag-response: Barranquilla", outer = TRUE)
```



Cartagena

The archive variables for each city during the study period are illustrated below:


Dengue Cases and variables climate of Cartagena
Cases
0
100
300
2010
2015
2020
Ave.temp.
25
27
29
2010
2015
2020
Spec.hum.
15
17
19
21
2010
2015
2020
Date
Rel.hum.
65
75
85
2010
2015
2020
Date
Total.prec.
0
50
100
2010
2015
2020
Date




```
cb_temp_ave_ctg<- crossbasis(
                  datos_dlnm$CTG_avg_temperature,
                  lag = 24,
                  argvar = list(fun = "ns", df = 3),
                  arglag = list(fun = "poly", degree = 2)
                )

cb_hum_rel_ctg<- crossbasis(
                  datos_dlnm$CTG_avg_relative_humidity,
                  lag = 24,
                  argvar = list(fun = "ns", df = 3),
                  arglag = list(fun = "poly", degree = 2)
                )

cb_prec_tot_ctg<- crossbasis(
                  datos_dlnm$CTG_total_precipitation,
                  lag = 24,
                  argvar = list(fun = "ns", df = 3),
                  arglag = list(fun = "poly", degree = 2)
                )
```

Now, with all terms parameterised, a negative binomial regression is fitted as follows:

```
mod_ctg <- glm.nb(Casos_Cartagena ~ cb_temp_ave_ctg+
                                    cb_hum_rel_ctg+
                                    cb_prec_tot_ctg+
                                     ns(Semana_Epidemiologica, df = 7) +
                                     factor(Año),
                                   data=datos_dlnm)
```

```
## Warning: glm.fit: algorithm did not converge
```

```
## Warning in glm.nb(Casos_Cartagena ~ cb_temp_ave_ctg + cb_hum_rel_ctg +
## cb_prec_tot_ctg + : alternation limit reached
```

```r
vif(mod_ctg)
```

```
##                                         GVIF Df GVIF^(1/(2*Df))
## cb_temp_ave_ctg                 3.955494e+05  9        2.046235
## cb_hum_rel_ctg                  1.689643e+08  9        2.864836
## cb_prec_tot_ctg                 1.130286e+07  9        2.465153
## ns(Semana_Epidemiologica, df = 7) 1.076872e+06  7        2.696925
## factor(Año)                     6.732241e+03 14        1.369998
```

```r
cptmean_ctg <- crosspred(
                basis = cb_temp_ave_ctg,
                model = mod_ctg,
                cen = mean(datos_dlnm$CTG_avg_temperature, na.rm = TRUE),
                by = 1)

cphrh_ctg <- crosspred(
              basis = cb_hum_rel_ctg,
              model =  mod_ctg,
              cen = mean(datos_dlnm$CTG_avg_relative_humidity, na.rm = TRUE),
              by = 0.5)

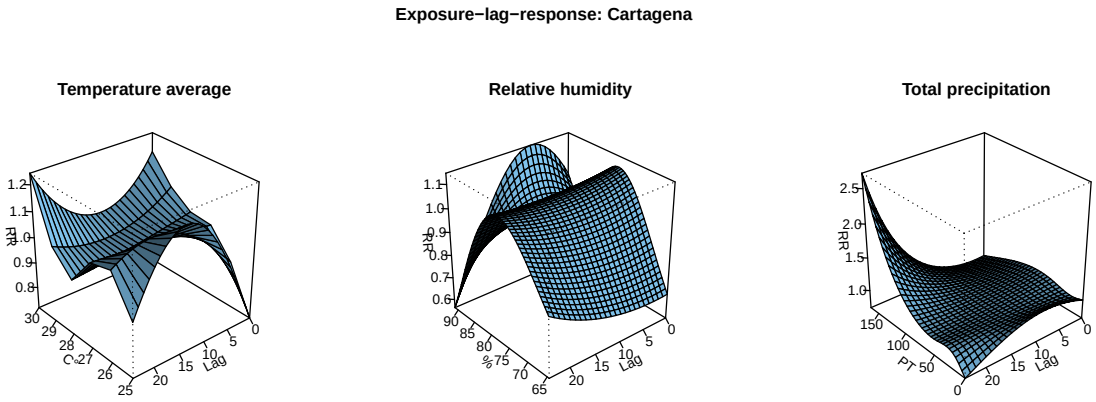
cpprecip_ctg <- crosspred(
                 basis = cb_prec_tot_ctg,
                 model = mod_ctg,
                 cen = mean(datos_dlnm$CTG_total_precipitation, na.rm = TRUE),
                 by = 5)
```

```r
par(oma = c(0, 0, 5, 0))

plot(cphrh_ctg,
     xlab = "%",
     zlab = "RR",
     main = "Relative humidity",
     phi = 30, theta = 230)

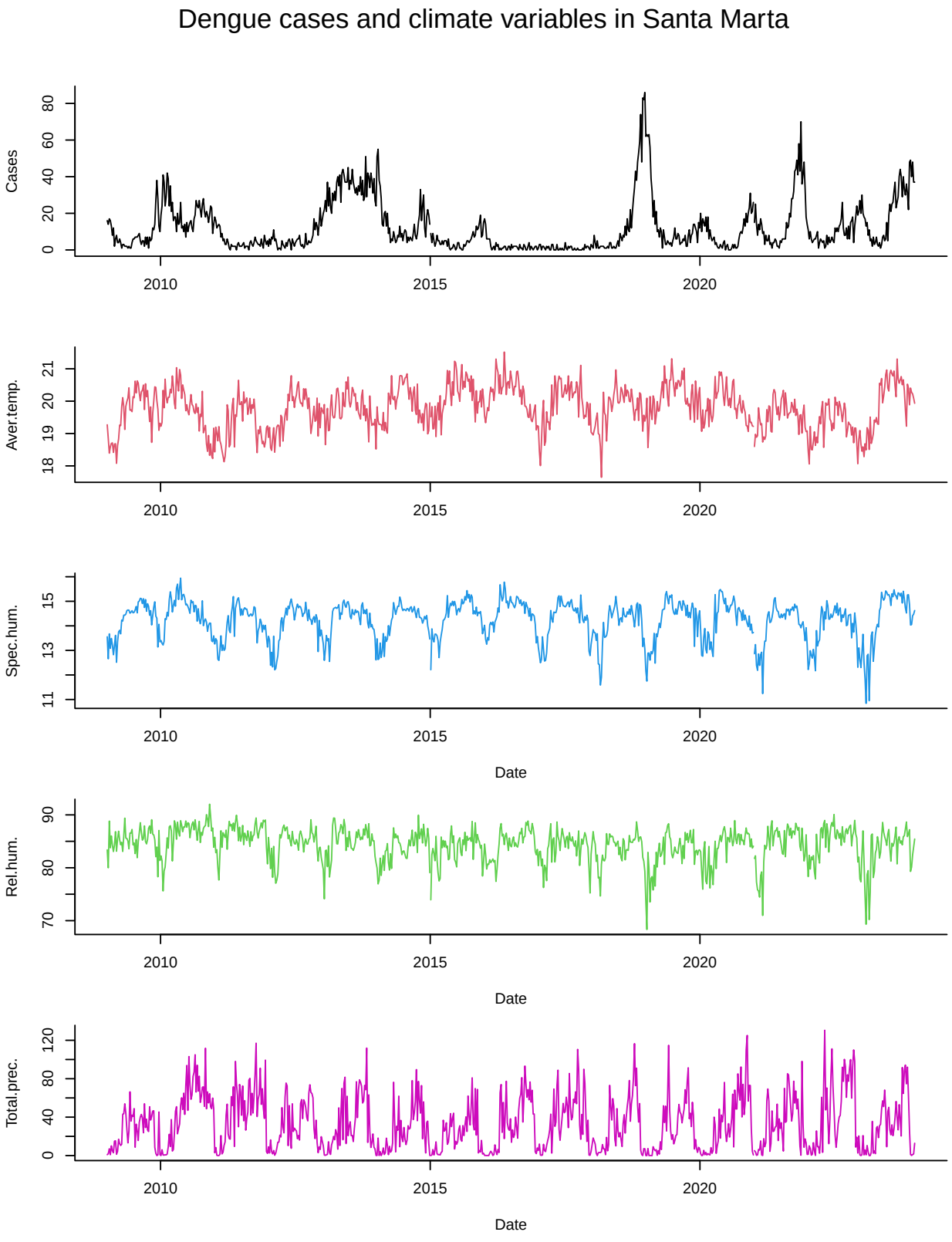
plot(cpprecip_ctg,
     xlab = "PT",
     zlab = "RR",
     main = "Total precipitation",
     phi = 30, theta = 230)

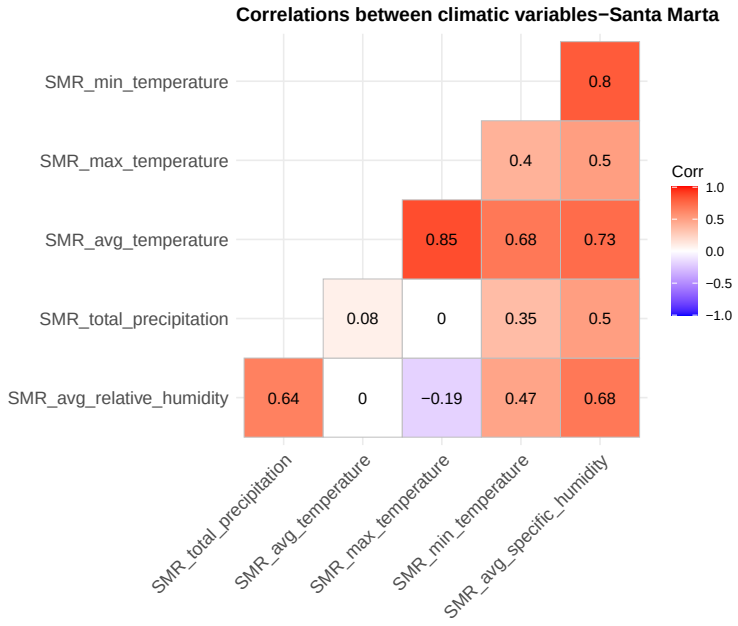
title(main = "Exposure-lag-response: Cartagena", outer = TRUE)
```



Santa Marta

The archive variables for each city during the study period are illustrated below:


Dengue cases and climate variables in Santa Marta
Cases
0
20
40
60
80
2010
2015
2020
Aver.temp.
18
19
20
21
2010
2015
2020
Spec.hum.
11
13
15
2010
2015
2020
Date
Rel.hum.
70
80
90
2010
2015
2020
Date
Total.prec.
0
40
80
120
2010
2015
2020
Date




```
cb_temp_ave_stm<- crossbasis(
                  datos_dlnm$SMR_avg_temperature,
                  lag = 24,
                  argvar = list(fun = "ns", df = 3),
                  arglag = list(fun = "poly", degree = 2)
                )

cb_hum_rel_stm<- crossbasis(
                  datos_dlnm$SMR_avg_relative_humidity,
                  lag = 24,
                  argvar = list(fun = "ns", df = 3),
                  arglag = list(fun = "poly", degree = 2)
                )

cb_prec_tot_stm<- crossbasis(
                  datos_dlnm$SMR_total_precipitation,
                  lag = 24,
                  argvar = list(fun = "ns", df = 3),
                  arglag = list(fun = "poly", degree = 2)
                )
```

Now, with all terms parameterised, a negative binomial regression is fitted as follows:

```
mod_stm <- glm.nb(Casos_SantaMarta ~ cb_temp_ave_stm+
                                     cb_hum_rel_stm+
                                     cb_prec_tot_stm+
                                      ns(Semana_Epidemiologica, df = 7) +
                                      factor(Año),
                                    data=datos_dlnm)
vif(mod_stm)
```

```
##                                       GVIF Df GVIF^(1/(2*Df))
## cb_temp_ave_stm                   38954.690  9        1.799000
## cb_hum_rel_stm                   266181.255  9        2.001699
```

```
## cb_prec_tot_stm                   8488603.675  9        2.426249
## ns(Semana_Epidemiologica, df = 7)  144226.134  7        2.336163
## factor(Año)                          6394.408 14        1.367481
```

```
cptmean_stm <- crosspred(
                basis = cb_temp_ave_stm,
                model = mod_stm,
                cen = mean(datos_dlnm$SMR_avg_temperature, na.rm = TRUE),
                by = 0.1)

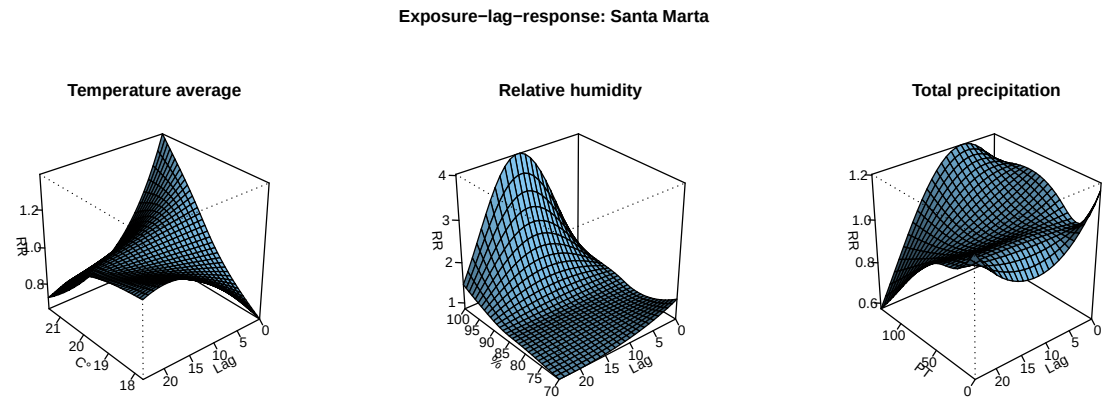
cphrh_stm  <- crosspred(
               basis = cb_hum_rel_stm,
               model = mod_stm,
               at = seq(70, 100, by = 1),
               cen = mean(datos_dlnm$SMR_avg_relative_humidity, na.rm = TRUE),
               by = 0.5 )

cpprecip_stm  <- crosspred(
                  basis = cb_prec_tot_stm,
                  model = mod_stm,
                  cen = mean(datos_dlnm$SMR_total_precipitation, na.rm = TRUE),
                  by = 5 )
```

```
par(oma = c(0, 0, 5, 0))

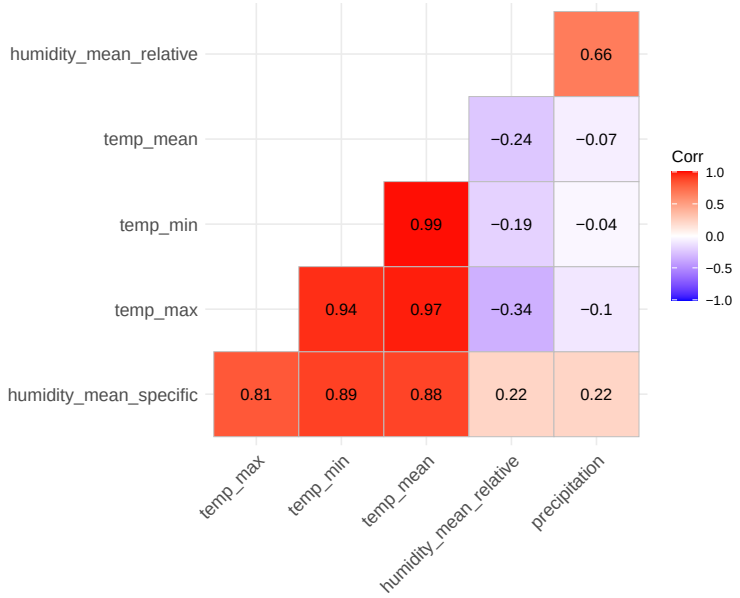
plot(cphrh_stm,
     xlab = "%",
     zlab = "RR",
     main = "Relative humidity",
     phi = 30, theta = 230)

plot(cpprecip_stm,
     xlab = "PT",
     zlab = "RR",
     main = "Total precipitation",
     phi = 30, theta = 230)
title(main = "Exposure-lag-response: Santa Marta", outer = TRUE)
```



Todas las ciudades juntas

For this, a base called 'data_final' was consolidated, where the variables were not separated by city, but the city was indicated in a new variable called 'city'.




```
cb_temp_mean <- crossbasis(data_final$temp_mean,
                           lag = 24,
                           argvar = list(fun = "ns", df = 3),
                           arglag = list(fun = "poly", degree = 2)
                           )

cb_humidity_relative <- crossbasis(data_final$humidity_mean_relative,
                                   lag = 24,
                                   argvar = list(fun = "ns", df = 3),
                                   arglag = list(fun = "poly", degree = 2)
                                   )

cb_precip <- crossbasis(data_final$precipitation,
                        lag = 24,
                        argvar = list(fun = "ns", df = 3),
                        arglag = list(fun = "poly", degree = 2)
                        )
```

```
model_bineg_1 <- glm.nb(dengue_cases ~ cb_temp_mean +
                                       cb_humidity_relative +
                                       cb_precip +
                                       ns(week, df = 7) +
                                       factor(year) +
                                       factor(city),
                        data = data_final)

vif(model_bineg_1)
```

```
##                              GVIF Df GVIF^(1/(2*Df))
```

```
## cb_temp_mean         435927.52583  9        2.057315
## cb_humidity_relative  65510.95450  9        1.851710
## cb_precip             12191.44492  9        1.686565
## ns(week, df = 7)       2301.16527  7        1.738358
## factor(year)             60.37069 14        1.157713
## factor(city)            305.06461  2        4.179246
```

```r
pred_temp_media <- crosspred(
                    basis = cb_temp_mean,
                    model = model_bineg_1,
                    cen = mean(data_final$temp_mean, na.rm = TRUE),
                    by = 1
                  )

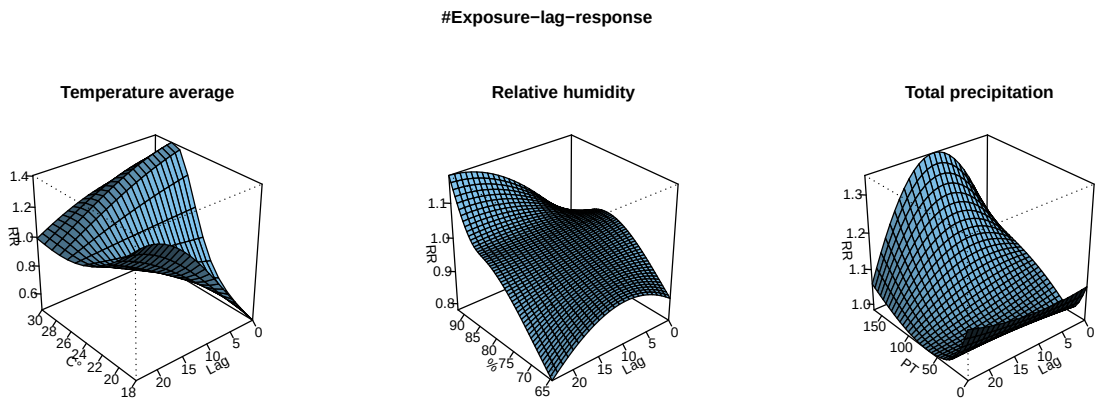
pred_humidity_mean_relative <- crosspred(
                               basis = cb_humidity_relative,
                               model = model_bineg_1,
                               cen = mean(data_final$humidity_mean_relative, na.rm = TRUE),
                               by = 0.5
                              )

pred_precipitation <- crosspred(
                        basis = cb_precip,
                        model = model_bineg_1,
                        cen = mean(data_final$precipitation, na.rm = TRUE),
                        by = 5
                      )
```

```r
layout(t(1:3))

plot(pred_precipitation,
     xlab = "PT",
     zlab = "RR",
     main = "Total precipitation",
     phi = 30, theta = 230)

title(main = "#Exposure-lag-response", outer = TRUE)
```



References